

Standardization of NaOH Titrant and Volumetric Determination of Unknown Acid

Prelab reading: Harris, *Quantitative Chemical Analysis*: Chapter 9

1. Learning Goals

- Review the basics of titrations and titration calculations
- Understand the meaning of a primary standard
- Determine the precision of your analysis
- Verify your quantitative laboratory skills

2. Introduction

In this experiment, potassium hydrogen phthalate ($\text{KHC}_8\text{H}_4\text{O}_4$, or KHP, $204.221 \text{ g mol}^{-1}$) is used as the primary standard for determining the exact concentration of your sodium hydroxide titrant. A primary standard is a chemical compound that is available in highly pure form (99.9+%), does not absorb water significantly, and is stable at high temperatures used for drying. KHP is a salt of phthalic acid ($\text{H}_2\text{C}_8\text{H}_4\text{O}_4$, or H_2P); only one proton per molecule of the KHP salt is acidic. The structures of the two molecules are:



In the first part of this laboratory exercise, you will prepare a solution that is approximately 0.10 M NaOH, and then determine its precise concentration by reacting it with potassium hydrogen phthalate (KHP), a primary analytical standard. Potassium hydrogen phthalate reacts with sodium hydroxide according to the equation:



In the second part of the laboratory, you will analyze a sample of a KHP unknown (a mixture of KHP with an inert salt such as NaCl) to determine the percent KHP in the samples. Both parts of the lab are based on the 1:1 reaction of NaOH with KHP illustrated in Fig. 1.

Hydrogen phthalate ion is a weak acid, so the initial solution will be acidic. At the equivalence point of the titration, all of the hydrogen phthalate ions are converted to phthalate ions, which are weakly basic ions, not neutral ions. Therefore, the pH at the equivalence point of the titration is somewhat above pH 7.0. This equivalence point corresponds closely to the pH at which phenolphthalein changes from its colorless form to its pink form, as illustrated in the right figure.

Atmospheric carbon dioxide, a weak acid when dissolved in water, can cause small but significant problems in acid-base titrations. In particular, CO_2 readily dissolves in basic solutions to form Na_2CO_3 or NaHCO_3 . You will need to take several measures to minimize its influence. First, you will prepare your NaOH titrant solution using a 10% by mass NaOH solution in which sodium carbonate is insoluble. All solution preparations will be carried out with deionized (DI) H_2O which has been boiled to eliminate any dissolved carbon dioxide. Finally, you should keep solutions in well-sealed containers as much as possible to minimize the introduction of atmospheric carbon dioxide.

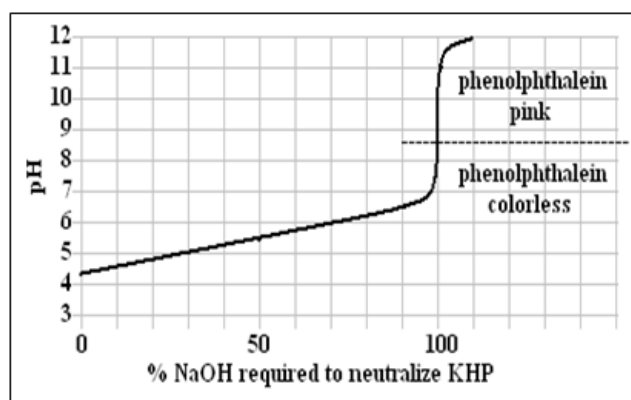


Figure 1: Sample titration curve

3. Reagents

- Potassium Hydrogen Phthalate (KHP, high purity)
- Phenolphthalein indicator solution
- 10% NaOH solution

4. Procedures

Note that this experiment will be carried out over two lab periods. In the first session, you will perform the standardization of NaOH. In the second session, you will titrate an unknown diprotic acid. In addition, you will be graded based on the quality of your data. If you do a poor job on your standardization of NaOH, you will not get good results for the diprotic acid titration.

Prepare standard potassium hydrogen phthalate (KHP) solution

- Primary standard grade KHP will be provided.
- Clean your flasks, beakers and volumetric pipettes with DI H₂O.
- Before coming to lab, calculate the amount of KHP required to prepare 250.0 mL of ~0.1 M KHP solution.
- Clean the flasks three times with 10~20 mL DI H₂O before use. If there are water droplets on the neck of the flask, it is not clean.
- Accurately weigh the KHP with analytical balance (use weigh boats). You should target ~5% accuracy for the mass.
- Add ~15 mL DI H₂O to a beaker, dissolve the KHP and then transfer to a 250.0 mL volumetric flask, rinse the beaker with 15 mL DI H₂O, transfer to the volumetric flask, repeat the rinsing procedure three times, and transfer all of the rinsed H₂O into the volumetric flask, then fill DI H₂O to the mark with pipet.
- Calculate the accurate concentration of KHP and write it on the flask and in your lab notebook.

Prepare NaOH solution

- ~10% NaOH solution will be provided.
- Before coming to lab, calculate how to prepare 500.0 mL 0.1 M NaOH solution from 10% NaOH solution.
- Use pipet to transfer desired ~10% NaOH solution to 500.0 mL volumetric flasks then fill DI H₂O to mark with pipet.
- You will use your standard KHP solution to standardize this solution.

Standardize the NaOH solution with standard KHP solution

- Set up a 50-mL burette, filling it with your NaOH solution.
- Before filling the burette, rinse it with a few small portions of the NaOH solution.
- Be sure to eliminate any bubbles from the burette's tip.
- Cover the top of the burette with a small piece of parafilm to prevent over-exposure to the atmosphere.
- Pipette 25.00 mL standard KHP solution to 250 mL Erlenmeyer flask.
- First Crude Titration: Do a fast crude titration to obtain an estimate of the volume of NaOH solution. Add 3 drops of phenolphthalein indicator to the flask, and titrate to the pink end point (This is just a rough end point). Record the volume of NaOH that was used for this rough titration in your lab notebook.
- Accurate Titration: Add 3 drops of phenolphthalein indicator to the flask, and titrate to the first faint pink end point that persists for 20 seconds. (Note that you are not looking for a dark pink solution. The light pink color should only persist for ~20 seconds, after that it should fade because of absorption of atmospheric CO₂).

- Titrate dropwise in the vicinity of the end point, and split drops at the end point. (Be sure to rinse down the sides of the flask.)
- You must obtain the same intensity of light pink color at the end point for all titrations.
- As you get closer to the endpoint of the titration, you should be adding the NaOH in ½-drops, and you should rinse down the sides of the flask with DI H₂O frequently.
- *Adding “½-drops” of titrant: Carefully open the burette’s stopcock so that a drop of titrant starts to form on the burette’s tip, and then close off the stopcock. Touch the burette’s tip to the side of the flask to transfer the partial drop to the flask, and rinse this down into the solution with deionized H₂O.*
- Perform at least three accurate titrations.
- Perform a blank titration with 25 mL of distilled water and subtract the value of the blank from each result.

5. Cleanup

- Clean and rinse all the glassware that you have used and place it on the drying rack. Place volumetric pipettes in the oven (careful: oven is hot!).
- Wipe your bench space clean.
- Dispose of all other wastes in the designated waste container.

6. Data Analysis and Presentation of Results in Lab Reports

NaOH standardization

- For each standardization titration, calculate the molarity of the NaOH titrant.
- Express the NaOH concentrations to at least four decimal places (e.g., 0.1023 M, or 0.0997 M).
- Calculate the average value, standard deviation, and RSD of your NaOH molarities.
- Be sure to report your final result along with the precision (95% confidence interval) in the form of $Interval = \bar{x} \pm \frac{ts}{\sqrt{n}}$. You will be graded mainly on the accuracy of your result. When you get t from the *Student’s t* table (Table 4-2), use $n-1$ as the degrees of freedom, where n is the number of measurements.
- If one result is quite different from the others, and there is no experimental reason to discard it, you may apply the Q test (at least 4 titrations needed for Q test). If this test rejects one of the molarity values, average the remaining values. It is always better to have more data; therefore run additional titrations as needed to obtain precise and accurate results.

Run	Volume of NaOH	Corrected Volume of NaOH	Concentration of NaOH	Average	SD	RSD
1						
2						
3						
4						
...						

Standardization of the pH electrode

Keep electrodes in aqueous solution when not in use.

The glass - bubble bottom of the electrode is very fragile. Do not touch or hit with stir bar during experiment.

1. Rinse electrode with DI water. Place in pH 7 standard solution (green-yellow)
2. Turn meter **on**
3. if the display doesn't show "pH" in upper left corner, press **pH** until it does
4. Press **calibrate** key
5. display should read 7.00. Press **Enter**
6. The 7.00 in the display will start to flash. May take a couple of minutes, but it should stop flashing and then the display will read 4.01 with 2 PT in the middle right.
7. Rinse electrode with more DI water. Place in pH 4 standard solution (pink)
8. Press **Enter**
9. The 4.01 in the display will start to flash. May take a couple of minutes, but it should stop flashing and then the display will read 4.00 ish – means the meter is calibrated. Don't turn off meter until class is over
10. If the pH display only reads to 0.1 pH unit, press the pH key twice – should now read to nearest 0.01 pH unit.
10. Rinse electrode and place back into water until ready for first reading.

To get pH reading – rinse off electrode with water. Place in solution. Don't record the number in the display until the "pH" in the upper left corner stops flashing (means you have a stable reading).